Digital Systems

Lecture 6

Subroutines

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Hilary Term 2025

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Following the course of past years by Mike Spivey

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These kinds of subroutines are very useful, because they are very **efficient**.

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Today we will discuss how to use RAM to solve these problems.

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Store registers in fixed memory addresses?

We cannot use absolute memory addresses

What about subroutines calling themselves (indirectly)?

Use a *Stack* for local variables

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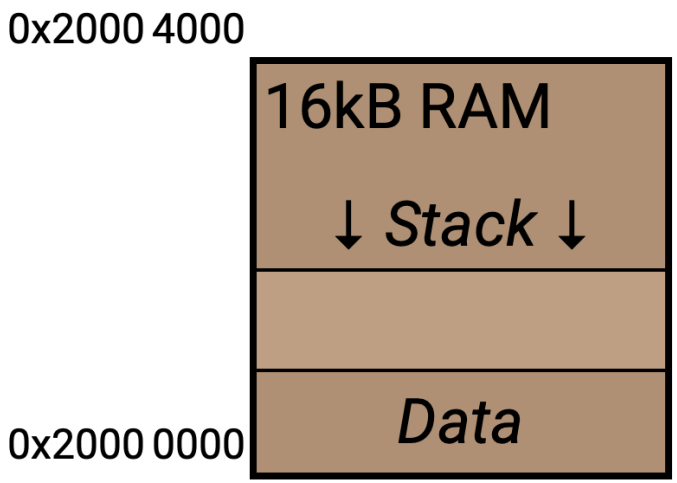
This is only one possible solution, but is the one chosen by design in ARM. Special instructions are created to facilitate using the stack.

The Stack



Example:

`main()`
`sub1()`
`sub2()`
`sub3()`

Return:

`main()`
`sub1()`
`sub2()`

Call next:

`main()`
`sub1()`
`sub2()`
`sub4()`

Call next:

`main()`
`sub1()`
`sub2()`
`sub4()`
`sub5()`

- Convention: Stack starts at top of address space, and grows downwards. So “above” in the stack has a lower address!
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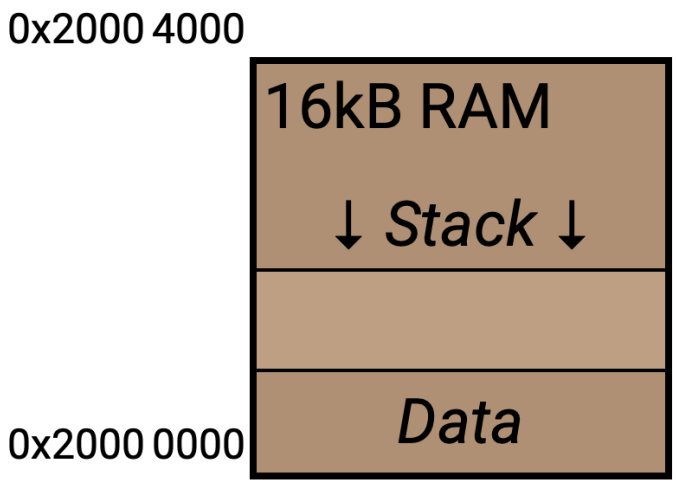

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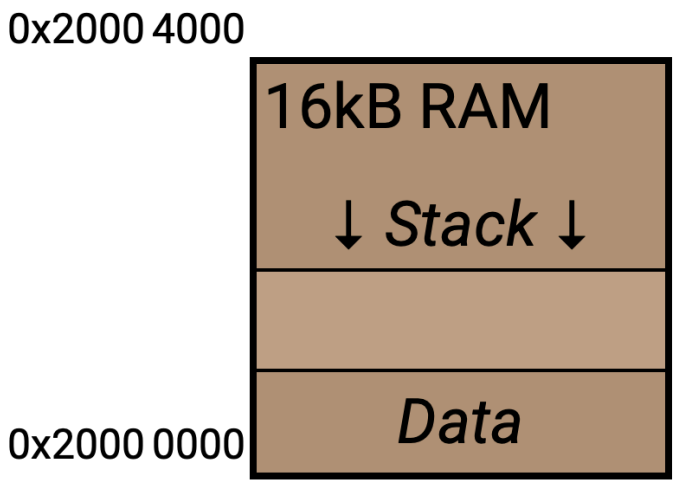

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- Convention: Stack starts at top of address space, and grows downwards. So "above" in the stack has a lower address!
- Each time a subroutine is called, `sp` points to a usable region of memory (stack frame), that won't interfere with other subroutines.

- When a subroutine returns, the caller finds its stack frame as it left it.

Subroutine Contract

Calling function:

- Places first arguments 1 … 4 in `r0` - `r3`
- Places arguments ≥ 4 on stack (called function knows how many to expect) (won't need this in course)

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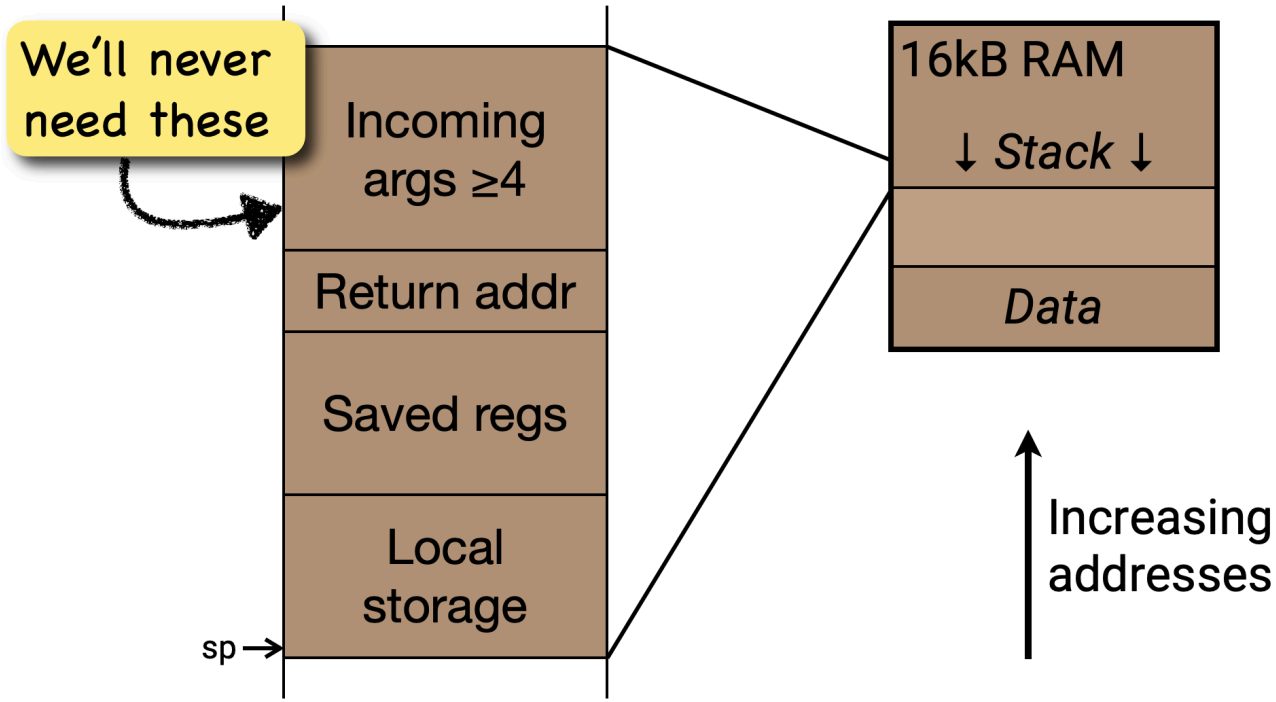
Called function:

- Leave values of `r4` and up unchanged on return
- Leave stack pointer `sp` unchanged on return
- Leave stack below current location (i.e. memory addresses above `sp`) unchanged
- Leave return value in `r0`

Subroutine Convention

Subroutine can use stack in whatever way it wants (assuming there is space!), but ARM provides instructions for:



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- `pop {REGISTER_LIST}`: Take values from stack, and place into the registers in list order, and increase `sp` accordingly.

```
func:
    push {r4, r5, lr} @ Save registers
    @ Use r4 and r5, call other routines
    pop {r4, r5, pc} @ Restore and return
```

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Normally, one instruction does one thing, but this gives speedupp! Load/store operation is 2 cycles, while `push`/`pop` is 1+n cycles.

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Remember, instruction set designers optimised the common case.

Putting it all together

Example: Factorial

```
int fac(int n) {
    int k = n, f = 1;
    while (k != 0) {
        f = mult(f, k);
        k = k-1;
    }
    return f;
}
```

Subroutine calls another subroutine! So we will use what we discussed.

Example: Factorial, opening

- As usual, let's start by converting code to assembly, and not worrying about efficiency.
- For clarity, I recommend making clear which variables are assigned to which registers (little table in your exam answer helps!)
- Let's assign `k` to `r4`, and `f` to `r5` (logic behind this?)
- Must use `push`.

```
fac:
    push {r4, r5, lr}
    movs r4, r0                 @ Initialise k to n
    movs r5, #1                 @ Set f to 1
```

Example: Factorial, Loop

Again, we can worry about optimisations later:

```
fac:
    push {r4, r5, lr}
    movs r4, r0                 @ Initialise k to n
    movs r5, #1                 @ Set f to 1
again:
    cmp r4, #0                  @ Is k = 0?
    beq finish                  @ If so, finished
```

Example: Factorial, Subroutine Call

Again, we can worry about optimisations later:

```
...
again:
    cmp r4, #0                  @ Is k = 0?
    beq finish                  @ If so, finished

    movs r0, r5                 @ Set f to f * k
    movs r1, r4
    bl mult                     @ branch link instruction!
    movs r5, r0
```

Example: Factorial, Finish Loop

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...
again:
    cmp r4, #0                  @ Is k = 0?
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    movs r0, r5                 @ Set f to f * k
    movs r1, r4
    bl mult                     @ branch link instruction!
    movs r5, r0

    subs r4, r4, #1             @ Decrement k
    b again                     @ And repeat
```

Example: Factorial, Return Subroutine

```
...
again:
    cmp r4, #0                  @ Is k = 0?
    beq finish                  @ If so, finished

...

finish:
    movs r0, r5                 @ Return f
    pop {r4, r5, pc}
```

Example: Factorial, All Together

```
fac:
    push {r4, r5, lr}
    movs r4, r0                 @ Initialise k to n
    movs r5, #1                 @ Set f to 1
again:
    cmp r4, #0                  @ Is k = 0?
    beq finish                  @ If so, finished

    movs r0, r5                 @ Set f to f * k
    movs r1, r4
    bl mult                     @ branch link instruction!
    movs r5, r0
```

```
    subs r4, r4, #1          @ Decrement k
    b again                  @ And repeat
finish:
    movs r0, r5              @ Return f
    pop {r4, r5, pc}
```

Example: Factorial, optimisation

You may have spotted that `f` can live in `r0` constantly, since the return value of `mult()` will also be the first argument for the next call.

Optimising compilers can spot this.

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- See notes for:
  - Expanded discussion
  - More efficient multiplication routine
- Don't forget rainbow chart and instruction overview!